

Rudra Arora

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SUMMARY

Third-year Computer Science co-op student at York University who builds software end to end - React frontends, Java and Python backends, REST APIs, relational schemas - containerized, tested and reviewed. Strongest on the data side of a product: ingestion, normalization and ETL over large financial datasets, and ML pipelines validated on held-out data rather than trusted. Following markets since 15 and investing personal capital across U.S., Canadian and Indian markets since 18; most of what I build sits where software and finance meet.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, HTML, CSS, C (intermediate), Verilog; object-oriented design and design patterns

Backend & Data: Spring Boot, FastAPI, Node.js/Express, RESTful API design and integration, PostgreSQL, SQLite, schema and query design, ETL and data pipeline development, point-in-time data modeling, concurrency

Frontend: React, HTMX, Jinja2, Streamlit, Plotly, REST API integration, server-side rendering, responsive UI

ML, AI & Quantitative: scikit-learn, LightGBM, PyTorch, pandas, NumPy, feature engineering, cross-validation and out-of-sample evaluation; financial statement analysis, valuation and profitability ratios; LLM API integration (Google Gemini), prompt engineering, tool/function calling

Tools & Practice: Git/GitHub (feature branches, pull requests, code review, Actions CI), Docker, pytest, JUnit, test-driven development, Agile/SDLC, technical documentation

EDUCATION

York University - Toronto, ON

Winter 2024 - Present

- Bachelor of Science (Hons.) in Computer Science, expected 2028; enrolled in York's Co-operative Education Program

PROJECTS

Financial Filings Data Platform (*Python, FastAPI, SQL, HTMX, Jinja2, pytest*)

- Built an end-to-end stock screener over public regulatory filings: ingestion, a normalization layer reconciling ~14,000 possible field names into one schema, and a query engine compiling user filters into parameterized SQL - 21 routes, 285,000 records, 85 tests.
- Kept the filing date on every datapoint so a historical screen returns only what was on file that day, instead of replaying today's restated figures back through history as free screeners do.
- Caught a trailing-twelve-month revenue figure running ~7% low because U.S. issuers never file Q4 standalone, and traced a bad per-share series to two incompatible source bases, recovering 27 corrections.

Job Market Salary Predictor (*Python, scikit-learn, LightGBM, FastAPI, PostgreSQL, pytest*)

- Engineered a feature pipeline over 19,000+ job postings - target encoding, role categorization, location parsing - then rebuilt it as a two-stage TF-IDF into gradient-boosting stack, raising accuracy from 54.6% to 79.8% against a 47.8% baseline.
- Compared RandomForest and HistGradientBoosting under stratified splits and 5-fold cross-validation, then validated on unseen employers to confirm it generalized rather than memorized; served it as a FastAPI microservice with a 57-test pytest suite.

AI-Powered Consultation Booking Platform (*React, Java, Spring Boot, PostgreSQL, Docker, Google Gemini API*)

- Built with a 3-person team across a design phase - class diagrams and interface contracts agreed first - and an implementation phase, in feature branches with code review before every merge to main.
- Modelled the appointment lifecycle as an enforced state machine, rejecting invalid transitions at the service layer instead of trusting frontend checks, and embedded a Gemini assistant inside the booking flow itself.

Full-Stack Chess Engine (*Java, Spring Boot, REST, JavaScript*)

- Built a server-side AI on minimax with alpha-beta pruning (depth 3), cutting search space ~70% versus brute force, with full rule validation server-side and concurrent sessions via ConcurrentHashMap.

MARKETS

Personal Portfolio Management - U.S., Canadian & Indian Markets

Ongoing

- Manage personal equity portfolios across three markets - a combined U.S./Canadian portfolio and a separate Indian portfolio - handling position sizing, currency exposure and differing market structures.
- Following markets since 15 and investing since 18; write the thesis down before entering a position and track it against benchmarks.

EXPERIENCE

Student Success Mentor Lead - YorkU Libraries

Fall 2024 - Present

- Lead 5 mentors across 3 campuses, standardizing scheduling and reporting workflows that contributed to a 30% increase in audience growth.
- Plan and execute 2 multi-campus events monthly, driving a 40% increase in sign-ups.

Study Space Assistant - YorkU Libraries

May 2026 - Present

- Support daily operations across 5 campus locations, coordinating logistics between staff and users.